Developing of Walkability Index for University Green Campus for the Evaluation of Existing Sustainable Pedestrian Walkability Infrastructure at PEC Campus in Chandigarh City for the Achieving of SDG-2030 Targets

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Abstract

In Developing countries like India, the walkability is an important concept in sustainable urban design approach. Sustainable urbanisation and technological advancements have significantly transformed city landscapes, emphasising the importance of sustainable living spaces in urban areas. The educational campuses, reflecting broader urban settings, are integral to this trend, with walkability index is developing as a key component. Walkability refers to how encouraging an area is to walking, influenced by factors such as pedestrian infrastructure, safety, accessibility, and environmental quality. This study examines the linear walkability index scores (LWI) of the Punjab Engineering College (PEC) campus, assessing its impact on students, faculty, and staff. The research aims to identify the strengths and weaknesses of the existing pedestrian infrastructure, understand campus users' perceptions, evaluate the environmental and aesthetic qualities of pedestrian spaces, and propose practical recommendations for improvements in university campus. Developing established frameworks like the Urban Walking Environment Context and the Pedestrian Level of Service (PLOS) model, the study employs surveys and observational methods to gather data. The findings reveal that while PEC boasts several strengths in existing pedestrian infrastructure, PEC Campus areas such as signage, aesthetic appeal, and sidewalk conditions need enhancement. The composite walkability score (CWS), calculated using the weighted scoring system, provided a quantitative summary of the overall walkability of the PEC campus. The final LWI score was 3.8 out of 5, indicating a generally positive but not perfect walkability environment. The study highlights the potential of campuses to serve as models for sustainable urban design, emphasising the sustainable integration of walkability index in PEC campus planning for the achieving of SDG-2030.

Keywords: Walkability, LWI, CWS, PEC green campus, Pedestrians, SDG-2030.

1. Introduction

1.1 Background and Context

Urbanization and technological advancements have dramatically transformed the landscapes of cities and towns around the world. One significant outcome of these changes is the increased focus on designing spaces that cater to sustainable living. Educational campuses, often seen as microcosms of broader urban settings, have not been left out of this trend. Among the many factors that contribute to the livability and sustainability of urban spaces, walkability stands out as a crucial element. Walkability refers to how friendly an area is to walking, influenced by factors such as infrastructure, safety, accessibility, and environmental quality. This concept has gained traction due to its implications for health, environmental sustainability, and social equity.

The Punjab Engineering College (PEC) occupies a significant amount of space and serves a large population of students, faculty, and staff. The design and functionality of its campus not only affect the daily lives of its inhabitants but also reflect the institution's commitment to sustainable practices. Fig. 1 illustrates the geographical layout of the PEC campus, providing a comprehensive view of its physical environment. Fig. 2 presents a digital map of the campus, highlighting key infrastructure and pathways. In this context, understanding the walkability of the PEC campus becomes a matter of interest and importance. This study explores the walkability infrastructure at PEC, assessing how it impacts various stakeholders and contributes to the overall campus experience.

Fig. 1. A geographical layout of the Punjab Engineering College campus, showcasing the overall physical environment and major landmarks.

Fig. 2. A digital representation of the PEC campus, highlighting key infrastructure, pathways, and important locations.

1.2 Objectives of the Study

The primary objective of this study is to evaluate the walkability of the PEC campus by examining the physical and environmental aspects of its pedestrian infrastructure. Specifically, the study aims to:

1. Identify the strengths and weaknesses of the existing walkability index infrastructure.
2. Modelling of LWI and understand the perceptions of campus users regarding the quality and adequacy of walking paths and related facilities in university.
3. Evaluate the environmental and aesthetic qualities of pedestrian spaces, including the presence of green cover and the overall ambience.
4. Propose recommendations for improving walkability, focusing on practical and implementable strategies that can be adopted by campus administrators and planners.

By achieving these objectives, the study seeks to bridge the gap between theoretical frameworks of walkability and real-world applications, offering insights that can inform future campus planning and policymaking for Indian cities and SDG-2030 Targets as per the United Nations in USA.

1.3 Significance of the Study

The significance of this study extends beyond the confines of the PEC campus. As a microcosm of broader urban settings, the campus offers valuable insights into the challenges and opportunities associated with promoting walkability. This research contributes to the academic literature on urban planning and sustainable transport, providing a case study that highlights the role of pedestrian-friendly infrastructure in enhancing campus life.

Moreover, the findings of this study have practical implications for campus administrators and policymakers. By providing a detailed evaluation of PEC's pedestrian infrastructure, the study identifies specific areas where improvements are needed. It also offers recommendations that align with broader sustainability goals, such as reducing carbon emissions and promoting healthy lifestyles. Ultimately, this research underscores the potential for campuses to serve as models of sustainable urban design, demonstrating how walkability can be integrated into the broader agenda of campus planning.

2. Literature Review

2.1 Conceptual Framework of Walkability

The concept of walkability encompasses a range of factors that contribute to the ease and attractiveness of walking as a mode of transport. Key components include connectivity, accessibility, safety, comfort, and aesthetics**.** Connectivity refers to the extent to which different parts of an area are linked by pedestrian paths, while accessibility focuses on the ease with which people can reach desired destinations on foot [1]. Safety involves both the actual and perceived risks associated with walking, such as the presence of well-maintained sidewalks and pedestrian crossings. Comfort and aesthetics pertain to the quality of the walking experience, influenced by factors such as sidewalk width, surface smoothness, and the presence of green spaces. Several theoretical models and frameworks have been developed to evaluate walkability. The Urban Walking Environment Framework and the Pedestrian Level of Service (PLOS) are notable examples. The Urban Walking Environment Framework emphasizes the importance of a holistic approach, considering both physical infrastructure and environmental quality. The PLOS model, on the other hand, focuses on quantifying the quality of pedestrian facilities based on criteria such as sidewalk width, traffic volume, and the availability of amenities. These models provide a foundation for assessing walkability in various settings, including educational campuses. A Review Paper: Study on Safety of Side Walkability Facilities in Urban Areas of Developing Countries [9].

2.2 Previous Studies on Campus Walkability

Research on campus walkability has gained momentum in recent years, reflecting a growing recognition of the importance of pedestrian-friendly infrastructure in academic settings. Studies conducted at universities worldwide have shown that enhanced walkability correlates with various positive outcomes, including higher levels of student engagement, better academic performance, and increased overall satisfaction with campus life. For instance, a study conducted at the University of California, Berkeley, found that improvements in pedestrian infrastructure, such as the addition of shaded walkways and seating areas, significantly enhanced the overall campus experience for students. Another study at the National University of Singapore highlighted the role of well-designed pedestrian paths in promoting physical activity among students, thereby contributing to their health and well-being. These studies underscore the multifaceted benefits of walkable campuses. They also highlight the diverse methodologies used to assess walkability, ranging from surveys and observational studies to geographic information system (GIS) analyses. By synthesizing findings from these studies, this section contextualizes the walkability of the PEC campus within a broader global framework, drawing comparisons and identifying best practices.

3. Research Methodology

3.1 Data Collection

The methodology for assessing the walkability of the Punjab Engineering College (PEC) campus is informed by two established models in urban planning and transportation studies: the Urban Walking Environment Framework and the Pedestrian Level of Service (PLOS) model. These frameworks provide a comprehensive basis for evaluating walkability by addressing both the physical and qualitative aspects of pedestrian environments.

Urban Walking Environment Framework: This framework emphasizes a holistic approach to assessing walkability, considering factors such as the presence and quality of sidewalks, street furniture, green cover, and the overall aesthetic and safety of the walking environment [2]. It ensures that various dimensions of the pedestrian experience are evaluated, not just the physical attributes.

Pedestrian Level of Service (PLOS): The PLOS model quantifies the quality of pedestrian facilities based on criteria such as sidewalk width, traffic volume, and the availability of amenities [3]. It provides a metric for evaluating the adequacy of pedestrian infrastructure, focusing on factors that directly impact pedestrian comfort and safety.

To align with these theoretical frameworks, we designed a comprehensive survey that captures both subjective perceptions and objective measurements of the walking environment at PEC. The survey targeted students, faculty, and staff, collecting data on various aspects, including:

Physical Infrastructure: Questions related to the condition, width, and connectivity of sidewalks, as well as the presence of crosswalks and signage.
Environmental Quality: Items assessing the presence of green cover, aesthetic appeal, and overall comfort and safety.
Safety and Security: Adequacy of lighting and presence of surveillance and security measures.
Accessibility and Inclusivity: Evaluating ramps, tactile paving, and other accommodations for people with disabilities, as well as proximity to essential campus facilities.
User Comfort and Experience: Subjective ratings of comfort and perceived safety.

3.2 Sample Characteristics for LWI

This section presents the demographic and descriptive characteristics of the sample population, as visualized through a series of bar charts. The data was collected from a survey conducted among students and faculty members at Punjab Engineering College for the size of 100 respondents from pedestrians.

The age distribution of respondents is illustrated in Fig. 3, with the majority (approximately 90%) falling within the "18-25" age group, reflecting the predominantly undergraduate population. A smaller proportion belongs to the "10-18" group, with negligible numbers in the "25-35" and "35+" categories. The gender distribution (see Fig. 4) shows an imbalance, with around 80% identifying as male and 20% as female, mirroring the gender ratio within the surveyed population. Regarding comfort levels while walking on sidewalks (see Fig. 5), about 80% of respondents reported feeling comfortable, while 20% expressed discomfort. The walking frequency (see Fig. 6) data reveals diverse habits, with most participants walking daily for varying durations, predominantly "1 hr - 2 hr," followed by "2 hr - 3 hr" and "4 hr +." This indicates a high level of pedestrian activity on campus.

In terms of preferences for green cover on sidewalks (see Fig. 7), an overwhelming majority (approximately 90%) expressed a positive preference, indicating a strong inclination towards natural and aesthetically pleasing environments. The perception of signage adequacy along sidewalks (see Fig. 8) was split, with about half of the respondents feeling that signage is sufficient, while the other half believed it was inadequate, suggesting a potential area for improvement. The aesthetic appeal of sidewalks (see Fig. 9) received mixed feedback; while a slight majority found them appealing, a notable portion rated them as unappealing, indicating room for enhancements. Accessibility to sidewalks (see Fig. 10) was generally rated as "Good" by most participants, reflecting effective infrastructure for pedestrian movement. However, the presence of crosswalks (see Fig. 11) was deemed sufficient by around 80%, though some indicated a need for more, especially in high-traffic areas.

Sidewalk conditions (see Fig. 12) were mostly rated as "Good," but a significant minority described them as "Bad," highlighting areas needing attention. The sufficient width for sidewalks (see Fig. 13) was affirmed by around 90% of respondents, although a small percentage disagreed, pointing to potential issues in crowded or narrow areas. Lastly, walking as a stress relief activity (see Fig. 14) was highly regarded, with approximately 95% of respondents finding it effective, underscoring the importance of pedestrian infrastructure in promoting mental well-being.
The sample characteristics reveal a diverse and engaged population with clear interests and concerns regarding pedestrian infrastructure. The findings provide valuable insights into the current state of sidewalks and highlight areas for potential improvement, aiming to enhance the safety, comfort, and aesthetic appeal of the walking experience on campus.

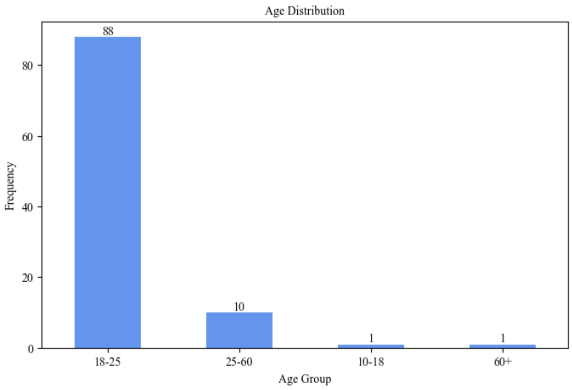

Fig. 3. Distribution of respondents by age group, with a majority in the "18-25" range.

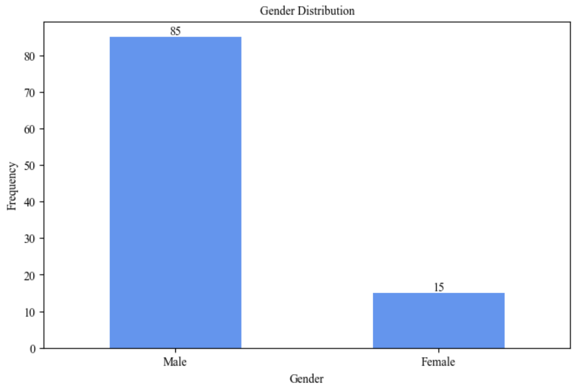

Fig. 4. Gender breakdown of survey respondents, predominantly male.

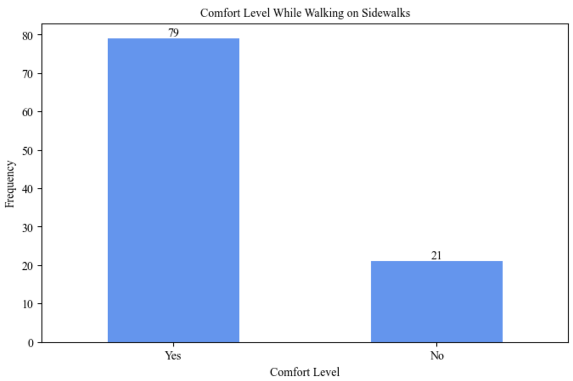

Fig. 5. Respondents' reported comfort levels when walking on sidewalks.

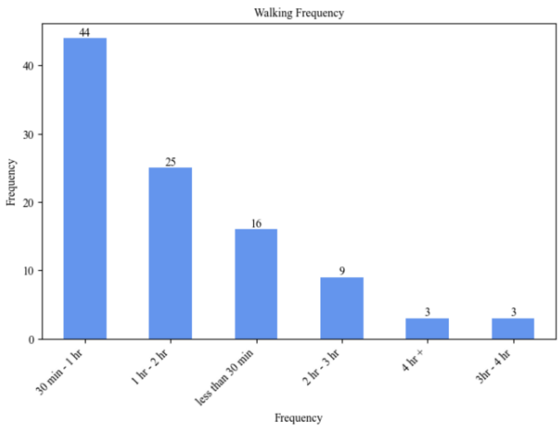

Fig. 6. Frequency and duration of walking among respondents.

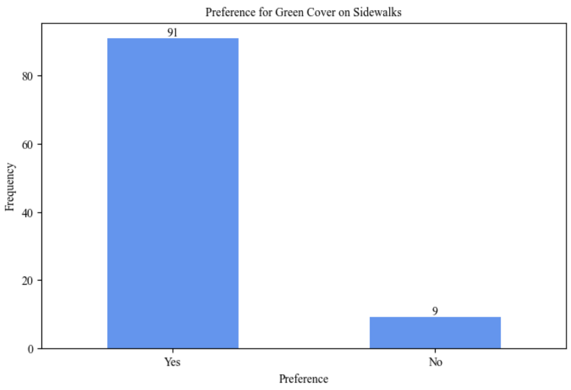

Fig. 7. Preference for green cover along sidewalks on campus.

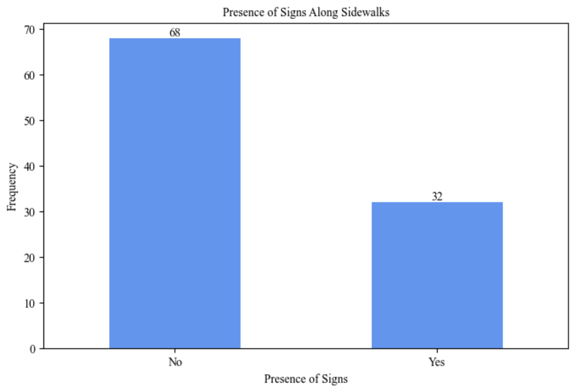

Fig. 8. Perception of signage adequacy along campus sidewalks.

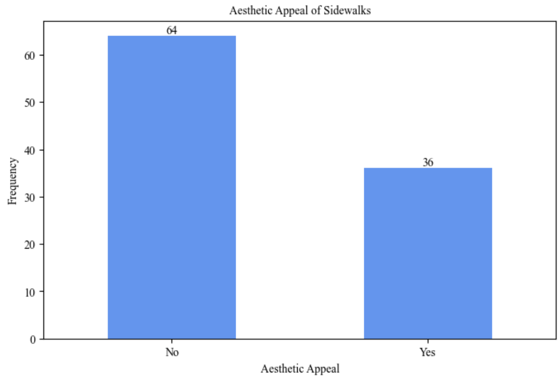

Fig. 9. Respondents' views on the aesthetic appeal of sidewalks.

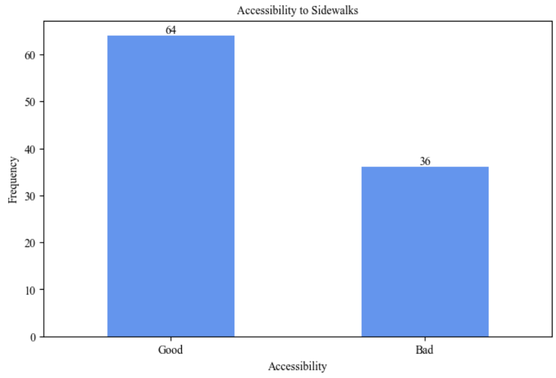

Fig. 10. Ratings of sidewalk accessibility near departments and market areas.

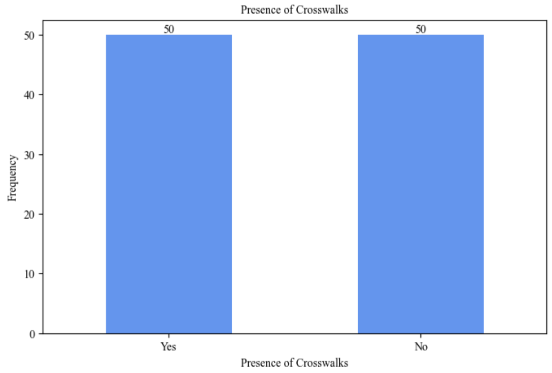

Fig. 11. Sufficiency of crosswalks throughout the campus.

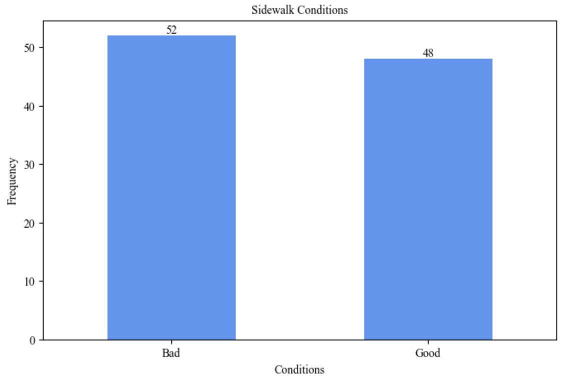

Fig. 12. Respondents' assessments of the condition of sidewalks.

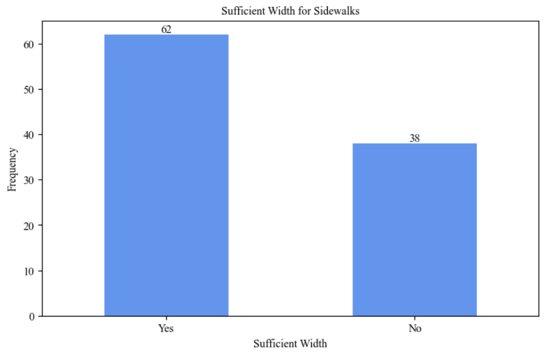

Fig. 13. Perceived adequacy of sidewalk width on campus.

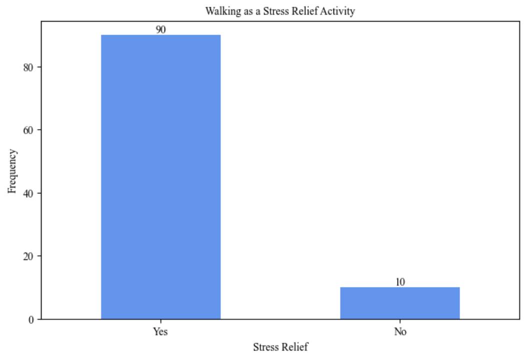

Fig. 14. The effectiveness of walking as a stress relief method for respondents.

3.3 Developing of Scoring and Weighting For LWI

To quantitatively assess the walkability of the PEC campus, a scoring system was established based on respondents' answers to the survey questions. Each factor was rated on a scale from 1 to 5, where higher scores indicated more favourable conditions. The scoring criteria for each factor were as follows:

Comfort: Scores were binary, with a "Yes" response scoring 5 and a "No" response scoring 1.
Frequency of Walking: Scored on a scale from 1 to 5, with 5 representing "4 hr +" and 1 representing "30 min - 1 hr".
Green Cover, Signage, Aesthetic Appeal, Crosswalk Availability, Sidewalk Width: Scored as 5 for "Yes" and 1 for "No".
Accessibility and Sidewalk Condition: Scored based on qualitative descriptions ranging from 5 (Excellent) to 1 (Very Poor).

This scoring system allowed for a comprehensive evaluation of the various components of walkability, providing a clear picture of the strengths and weaknesses of the campus's pedestrian infrastructure. The survey responses were quantified using a scoring system that reflects the principles of the Urban Walking Environment Framework and PLOS. Each factor was scored on a scale from 1 to 5, with higher scores indicating better conditions or more favourable perceptions [4]. The scores were then weighted based on relative importance, as informed by literature and expert judgment.

Table 1. Assigned weights for each factor.

Factor	Weight
Sidewalk Width and Condition (combined)	0.20
Crosswalk Availability and Safety	0.15
Green Cover	0.10
Aesthetic Appeal	0.05
Lighting	0.10
Surveillance and Security	0.10
Accessibility for Disabled Individuals	0.10
Proximity to Amenities	0.10
Comfort and Perceived Safety	0.10

3.4 Modelling of Linear Walkability Index

First model: The overall walkability index was calculated using the formula as shown in Equation 1 and 2.

$$\text{Linear Walkability Index} = \sum (\text{Score}_i \times \text{Weight}_i) \quad (1)$$

$$LWI = \sum (S_i \times W_i) \quad (2)$$

Where: S_i - Score_i represents the score for each factor and W_i - Weight_i represents the corresponding weight [5].

Second model: The Ministry of Urban Development (MoUD) established this system under the Government of India. They specifically developed this strategy based on the condition of Indian roadways. According to this technique, walkability records are a component of pathway accessibility and person-on-foot office rating.

Walkability Index = [(W1 x Availability of footpath) + (W2 x Pedestrian rating)]

W1 and W2 is assumed to be 50% for Indian roads and conditions
Availability of footpath length in the study area = Footpath length in study area / Length of roads in the city
Pedestrian Rating = Local pedestrians opinion based on different parameters

The main disadvantage of this strategy is the difficulty in identifying the parameters that need to be improved, such as comfort, security, and safety. This strategy is only applicable to walkways with widths greater than 1.2m. The length of main roads and walkways is determined by roadometers.

Pedestrian’s facilities include various parameters like:

- Width of foothpath
- Continuity of foothpath
- Crossing availability
- Cleanliness and maintenance
- Foothpath surface
- Obstructions on foothpath
- Amenities

Using these parameters pedestrians’ facility rating is calculated on the scale of 5.

3.5 Video camera data for PEC green campus

The data images were collected from different stretches of sidewalk at PEC Campus as shown in Fig.15

Fig .15. Details of different stretches images of sidewalk in PEC Campus, Chandigarh city, India.

4. Model Results and Analysis

The following LWI model results were discussed in Table 2 and Table 3.

4.1 Model Survey Results

The survey received responses from a diverse group of campus users, providing a comprehensive overview of the walkability conditions at PEC. The analysis revealed a range of insights into the various factors influencing walkability.

Table 2. Key findings for each factor affecting walkability.

Factor	Key Findings
Comfort	79% feel comfortable; concerns about overcrowding and obstacles in high-traffic areas (see Fig. 16).
Frequency of Walking	Over 60% walk at least 2 hours daily, indicating high pedestrian activity.
Green Cover	Over 80% prefer sidewalks surrounded by greenery, highlighting the importance of aesthetics and environment.
Signage	60% find signage adequate, but some suggest more signs, especially for first-time visitors.
Aesthetic Appeal	55% rate the sidewalks as appealing; some note issues like uneven surfaces and lack of greenery in certain areas.
Accessibility	75% find sidewalks accessible from major campus locations; some suggest improvements for those with disabilities.
Crosswalk Availability	85% indicate sufficient crosswalks, but concerns exist about their visibility and maintenance.
Sidewalk Condition	65% rate sidewalk condition as good; some areas need repair.
Sidewalk Width	70% feel sidewalks are sufficiently wide, but congestion noted during peak hours.

These results provide a comprehensive snapshot of the current state of walkability at the PEC campus, highlighting both strengths and areas for improvement.

4.2 Model of LWI Observational Data

The observational studies corroborated many of the survey findings, providing additional detail on specific aspects of the pedestrian infrastructure.

Sidewalk Width and Condition: Measurements indicated that the majority of sidewalks met the recommended width standards, though some narrower paths were identified. The condition of the sidewalks varied, with some sections in need of resurfacing and repair (see Fig. 17).

Green Cover and Aesthetics: The presence of green cover was uneven, with some areas boasting lush greenery and others relatively barren. The aesthetic quality of the sidewalks was also variable, with well-maintained sections contrasting with areas suffering from neglect (see Fig. 17).
Crosswalks and Signage: Observations confirmed the adequate presence of crosswalks, though some were found to lack proper signage and markings. The quality and clarity of signage varied, with some signs being outdated or difficult to read (see Fig. 17).

Additionally, the survey data revealed significant variations in walkability scores across different age groups, as depicted in the heatmap (see Fig. 15). The heatmap indicates that the age group 10-18 and 60+ reported higher walkability scores compared to other age groups, with scores of 4.2 and 4.3 respectively. In contrast, the age groups 18-25 and 25-60 reported lower walkability scores, with scores of 3.1 and 2.8 respectively. This variation suggests that different age groups perceive the walkability of the campus differently, possibly due to varying physical abilities and expectations. These observations provided a more detailed understanding of the physical characteristics of the campus's pedestrian infrastructure, complementing the subjective survey data.

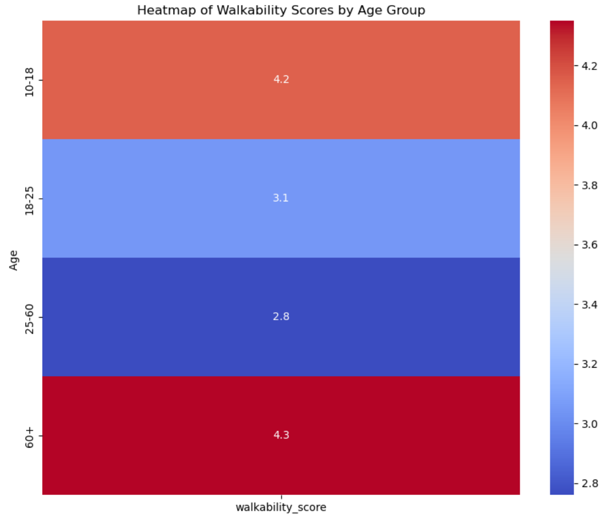

Fig. 16. A heatmap displaying the average walkability scores across different age groups, highlighting variations in perceived walkability based on age.

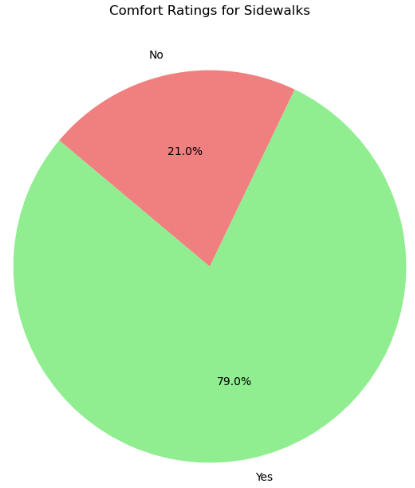

Fig. 17. A pie chart depicting the distribution of comfort ratings for sidewalks, indicating the percentage of respondents who feel comfortable versus those who do not.

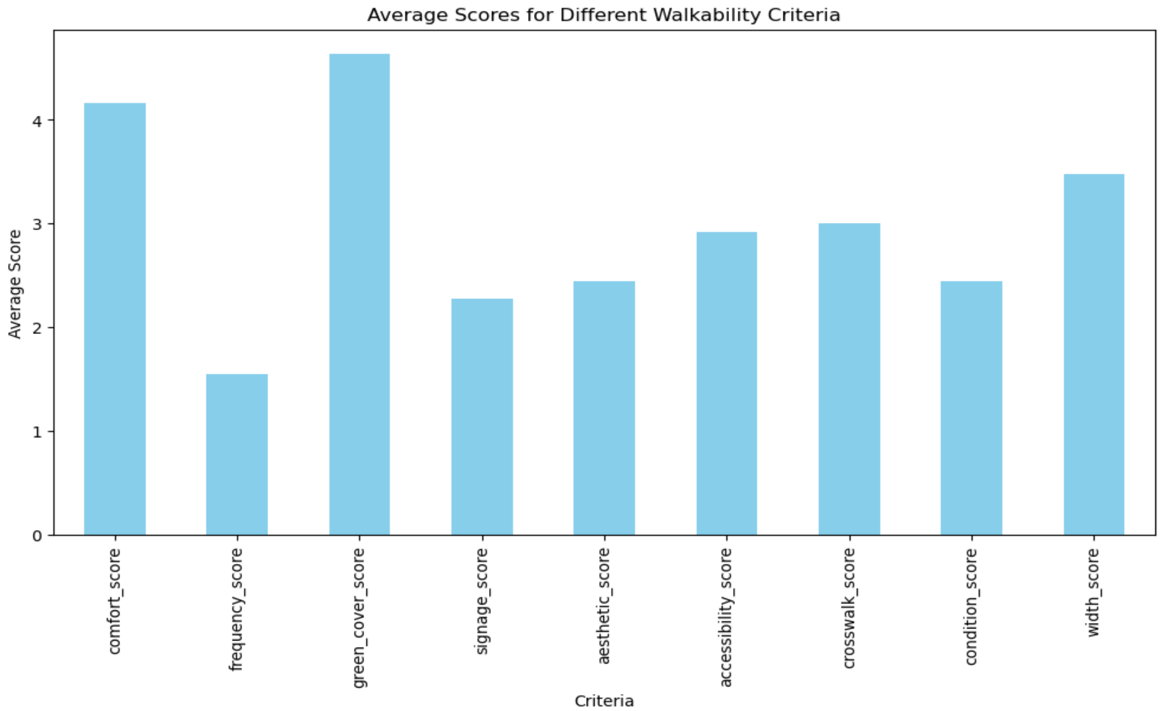

Fig. 18. A bar chart illustrating the average scores for various walkability criteria, including comfort, frequency, green cover, signage, aesthetic appeal, accessibility, crosswalk presence, sidewalk condition, and width.

4.3 Composite Walkability Score

The composite walkability score (CWS), calculated using the weighted scoring system, provided a quantitative summary of the overall walkability of the PEC campus. The final score was 3.8 out of 5, indicating a generally positive but not perfect walkability environment and as shown in Table 3.

Table 3. Walkability index (LWI) score for each factor affecting walkability.

Factor	Score
Comfort	4.0
Frequency of Walking	3.5
Green Cover	3.8
Signage	3.2
Aesthetic Appeal	3.5
Accessibility	4.0
Crosswalk Availability	4.3
Sidewalk Condition	3.7
Sidewalk Width	3.9

These scores highlight the strengths of the campus's walkability infrastructure, such as comfort and accessibility, while also pointing to areas where improvements are needed, such as signage and aesthetic appeal [6].

5. Discussion

5.1 Strengths of PEC's Walkability Infrastructure

The PEC campus boasts several strengths in terms of walkability. The high level of comfort reported by respondents indicates that the majority of the campus's pedestrian infrastructure meets users' needs. The widespread presence of crosswalks and the generally good condition of sidewalks further enhance the safety and convenience of walking [7]. The preference for green cover and the importance placed on aesthetics by respondents underscore the value of environmental and visual quality in creating a pleasant walking experience.

5.2 Areas for Improvement

Despite these strengths, the study identified several areas where improvements could be made. The mixed reviews on signage and aesthetic appeal suggest that more attention could be paid to enhancing the clarity and attractiveness of pedestrian spaces. The variability in sidewalk conditions and the presence of narrower paths in some areas indicate the need for targeted maintenance and upgrades.

5.3 Comparison with Other Campuses

Comparing the PEC campus with other academic institutions reveals both common challenges and unique features. Like many campuses, PEC faces the challenge of balancing pedestrian and vehicular traffic, particularly in busy areas. However, the campus's relatively high score for crosswalk availability suggests that it may be doing better than some peers in ensuring safe pedestrian crossings. The emphasis on green cover and aesthetics also aligns with broader trends in campus planning, which increasingly prioritize environmental sustainability and quality of life.

6. Recommendations

6.1 Enhancing Green Cover and Aesthetic Appeal

To improve the overall walking experience, the campus could invest in increasing green cover along pedestrian paths. This could involve planting additional trees, shrubs, and flowers, as well as creating green corridors that provide shade and improve air quality. Enhancing the aesthetic appeal of sidewalks through better maintenance, the addition of public art, and improved lighting could also make walking more enjoyable.

6.2 Improving Signage and Wayfinding

Upgrading the campus's signage system is another critical area for improvement [8]. This could include installing more visible and informative signs, particularly at key intersections and entry points. The use of digital displays and interactive maps could further aid in wayfinding, making it easier for visitors and new students to navigate the campus.

6.3 Addressing Sidewalk Condition and Width

Regular maintenance of sidewalks is essential to ensure safety and comfort. The campus could implement a systematic inspection and repair program to address issues such as cracks, uneven surfaces, and drainage problems. In areas with high pedestrian traffic, widening sidewalks could help alleviate congestion and improve the overall flow of foot traffic.

6.4 Promoting a Pedestrian-Friendly Culture

In addition to physical infrastructure improvements, the campus could promote a pedestrian-friendly culture through initiatives such as walking clubs, awareness campaigns, and events that celebrate walking as a healthy and sustainable mode of transport. Encouraging the use of public transport and cycling could also help reduce reliance on cars, further enhancing the walkability of the campus.

7. Conclusion

This study offers an in-depth evaluation of the walkability infrastructure at Punjab Engineering College (PEC), highlighting both the institution's strengths and areas that require improvement. The analysis underscores the crucial role pedestrian infrastructure plays in shaping the everyday experiences of students, faculty, and staff, as well as in promoting sustainable urban development within academic environments. Among the key strengths identified are comfort and accessibility, with a significant proportion of respondents expressing satisfaction with their walking experience on campus. The campus's thoughtful design facilitates seamless movement between academic buildings, residential areas, and other essential facilities, making it easy for individuals to navigate the space on foot. One of the standout features is the availability and safety of crosswalks. The campus is well-equipped with adequately marked crosswalks, particularly in high-traffic areas, which are crucial for ensuring pedestrian safety. These crosswalks help mitigate potential conflicts between pedestrians and vehicular traffic, thereby creating a safer and more user-friendly environment. Additionally, the presence of green cover, though uneven across the campus, contributes positively to the overall walking experience.

The existing greenery not only enhances the aesthetic value of the campus but also provides environmental benefits, such as shade and improved air quality, which are essential for a comfortable and pleasant walking experience. Despite these strengths, the study identifies several critical areas where improvement is necessary. One major area is signage and wayfinding. The feedback indicates a need for clearer and more informative signage, particularly at key intersections and entry points. Enhancing the visibility and informativeness of signs can significantly improve navigability, making it easier for both new and returning campus users to find their way. Another area for improvement is the aesthetic appeal and condition of sidewalks. While some parts of the campus are visually appealing, others are less maintained. Addressing uneven surfaces, ensuring pathways are clean and well-lit, and incorporating more aesthetic elements such as public art can significantly enhance the visual experience and overall satisfaction of pedestrians. Furthermore, the condition of the campus's infrastructure, particularly sidewalks, requires consistent maintenance. It is essential to ensure that sidewalks are free of obstructions and have adequate width to accommodate pedestrian traffic comfortably. This is particularly important in crowded areas, where the flow of pedestrian traffic can be hindered by narrow or poorly maintained pathways. Additionally, the study highlights the need for greater inclusivity in campus infrastructure. Although the campus is generally accessible, there are opportunities to enhance accessibility further by adding accommodations for individuals with disabilities.

This includes implementing tactile paving and ramps to ensure that all users, regardless of physical ability, can navigate the campus safely and comfortably. The insights from this study are not only valuable for PEC but also for other academic institutions aiming to improve their walkability. As campuses increasingly focus on promoting sustainable practices, enhancing walkability becomes a critical component of campus planning. The findings suggest that targeted interventions in areas such as signage, aesthetic appeal, and infrastructure maintenance can significantly improve the pedestrian experience. Future research should continue exploring the relationship between campus design and user satisfaction, particularly concerning sustainable development. Longitudinal studies assessing the impact of the recommended improvements on walkability and campus life could provide further valuable insights. Moreover, expanding the scope of research to include diverse campuses with varying geographical and cultural contexts could help in understanding the broader applicability of these findings. All in all, this study emphasizes the vital importance of walkability in creating vibrant, sustainable, and inclusive campus environments. By addressing the identified challenges and leveraging the strengths of its existing infrastructure, PEC has the opportunity to set a benchmark for other institutions. This would not only enhance the quality of life for its current community members but also position PEC as a leader in sustainable campus design, demonstrating the significant benefits of prioritizing pedestrian-friendly infrastructure as per the SDG-2030 targets and recommendations.

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Appendix: Online Google Survey Form


Questions Responses 100 Settings

Sidewalks at PEC

Form description

This form is automatically collecting emails from all respondents. Change settings

We are students from Punjab Engineering College (PEC), and we are conducting a study to understand the preferences of stakeholders regarding the sidewalks **within the PEC campus**. Your input will be invaluable in helping us assess current conditions and identify potential improvements.

PEC
EXPLORE INNOVATE EXCEL
SINCE 1921
CELEBRATING
100 YEARS

(https://docs.google.com/forms/d/e/1FAIpQLSdYbrcdwO3Fpier7ROW-4m5IVfFfisczq_Gk-dlvJ-4uCBztA/viewform?usp=sf_link)

<u>Example of data set for Online Google Survey Form</u>

Question 1. Age
Question 2. Gender
Question 3. Do you feel **comfortable** while walking on sidewalks?
Question 4. How **frequently** do you walk?
Question 5. Do you prefer sidewalks surrounded by **green cover** on the PEC campus?
Question 6. Are there enough **signs along the sidewalks** for different locations within the college?
Question 7. Is there **shade** on the sidewalks during the day?
Question 8. Ample **lighting** during the night?
Question 9. Do you **feel safe**, walking on sidewalks?
Question 10. Would you say that sidewalks are **aesthetically appealing** for walking on?
Question 11. **Accessibility** to sidewalks directly outside departments and the market?
Question 12. **Sufficient presence** of crosswalks throughout campus?
Question 13. Sidewalk **conditions**?
Question 14. Sufficient **width** for the sidewalk?
Question 15. Does walking help you with **stress relief**?